Credit card fraud: timing and payment patterns

A two-day transaction study | Timing, repeated amounts and concentration of fraud-labelled value

All transactions

284,807

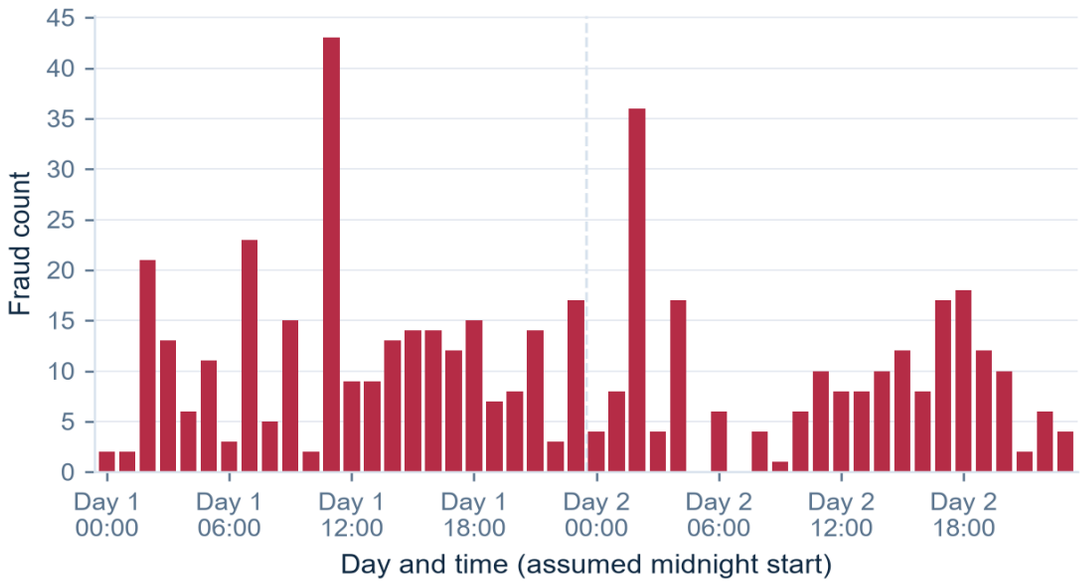
Confirmed fraud

492

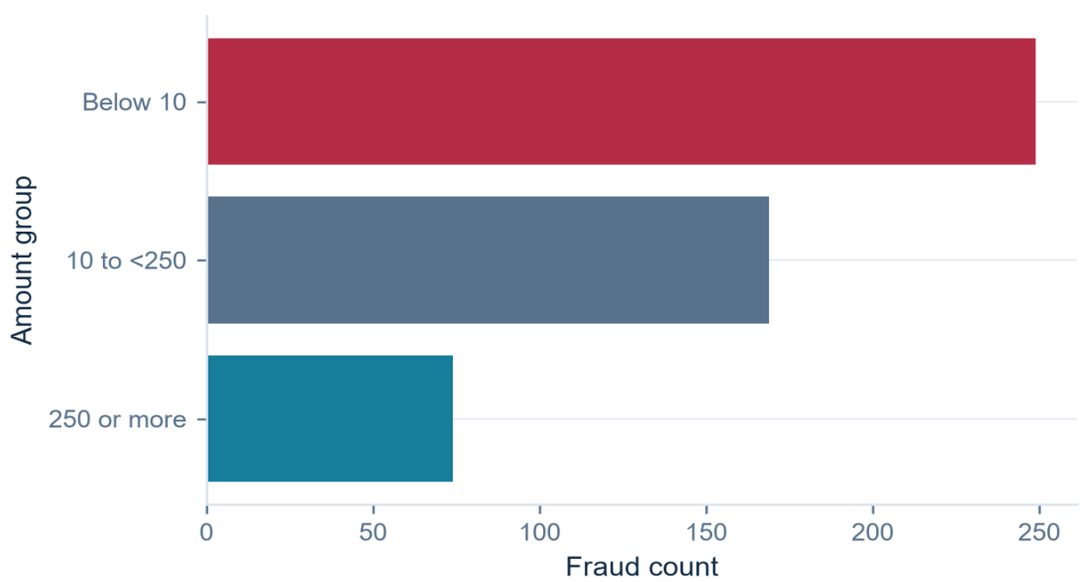
Amount in fraud transactions

60,127.97

Fraud transactions by hour



Fraud transactions by amount



Half of the fraud payments were below 10 (249 of 492). Payments of 250 or more were less common, but accounted for 73.9% of fraud-labelled value. Under the midnight-start assumption, the highest hourly fraud rate on both days was between 02:00 and 03:00.

Time assumption: Day 1 starts at 00:00; calendar anchor 01 Jan 2000 is illustrative. No real source date is available.

Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

The recording window covers almost exactly two days

Time is seconds since the first transaction. Midnight is an analysis assumption requested for this report.

Elapsed seconds	Assumed timestamp	How it reads
0	01 Jan 2000, 00:00:00	Day 1 begins
39,600	01 Jan 2000, 11:00:00	11:00 on Day 1
86,400	02 Jan 2000, 00:00:00	Day 2 begins
93,600	02 Jan 2000, 02:00:00	02:00 on Day 2
172,792	02 Jan 2000, 23:59:52	Final recorded transaction

Observation span

47h 59m 52s

Missing cells

0

Identical extra rows retained

1,081

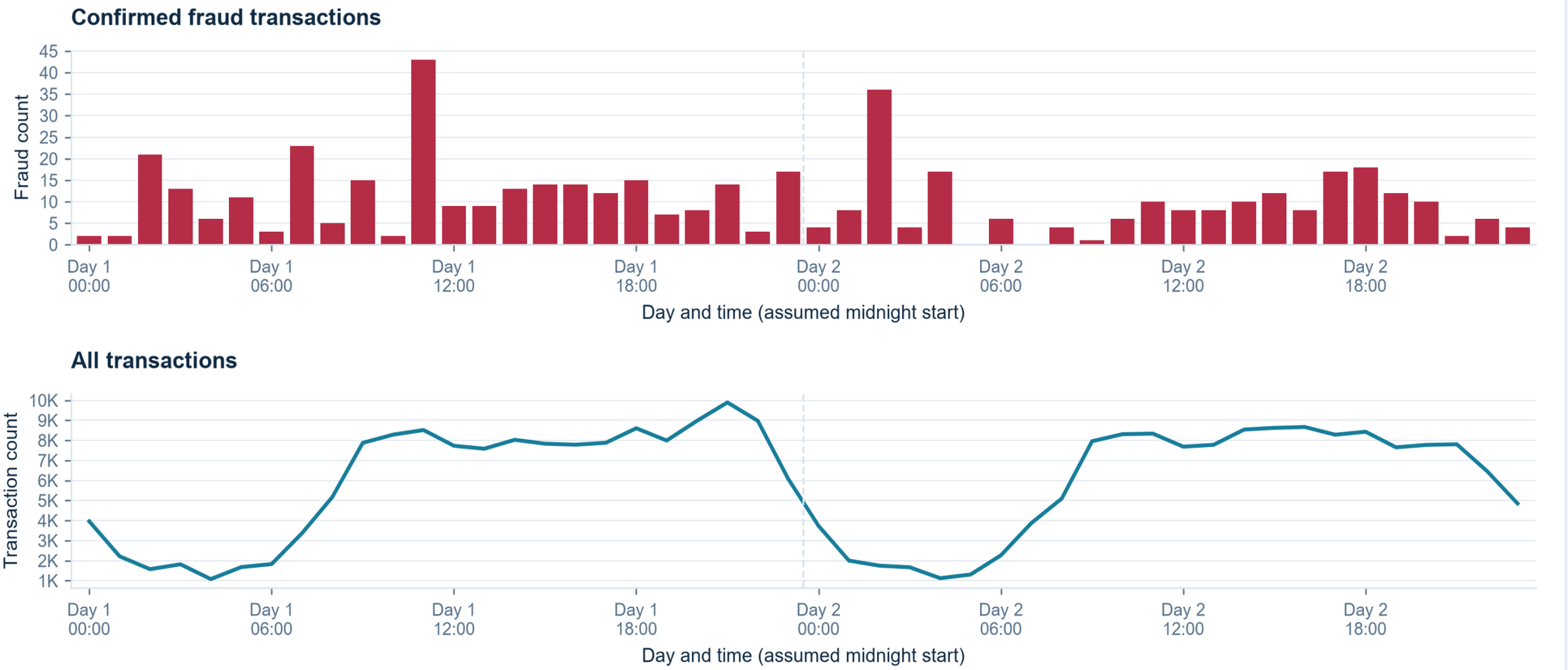
The anchor date makes time labels easier to read; it is not the bank's actual transaction date. V1-V28 are anonymised features with no supplied business meaning. The main analysis retains every row. A separate gap sensitivity view collapses matching fraud rows with the same second and amount; this does not prove those rows are duplicates.

Time assumption: Day 1 starts at 00:00; calendar anchor 01 Jan 2000 is illustrative. No real source date is available.

Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

The busiest fraud hour was 11:00 on Day 1

Read fraud counts alongside all payment activity. More fraud cases can simply coincide with more payments.



Day 1, 11:00-12:00 contained 43 confirmed fraud payments among 8,517 transactions. That was the largest fraud count in any hour, but its fraud rate was 0.50%. The hours with the highest fraud rates had much lower total activity.

Time assumption: Day 1 starts at 00:00; calendar anchor 01 Jan 2000 is illustrative. No real source date is available.

Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

Fraud was more common at 02:00 on both days

Fraud rate = confirmed fraud payments / all payments in the same hour. The denominator matters.

Overall fraud rate

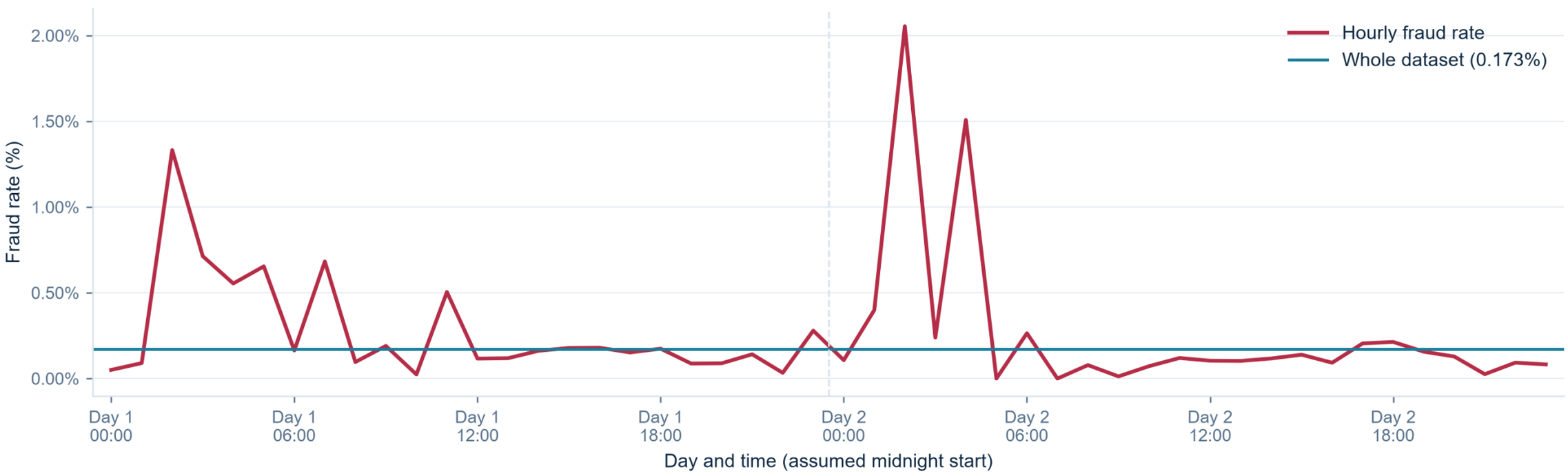
0.173%

Day 1, 02:00-03:00

1.33%

Day 2, 02:00-03:00

2.05%



The Day 2 peak was 36 fraud payments out of 1,752 transactions (2.05%), compared with 21 out of 1,576 (1.33%) at the same hour on Day 1. This is a useful review window in these two days; it does not establish a lasting overnight pattern.

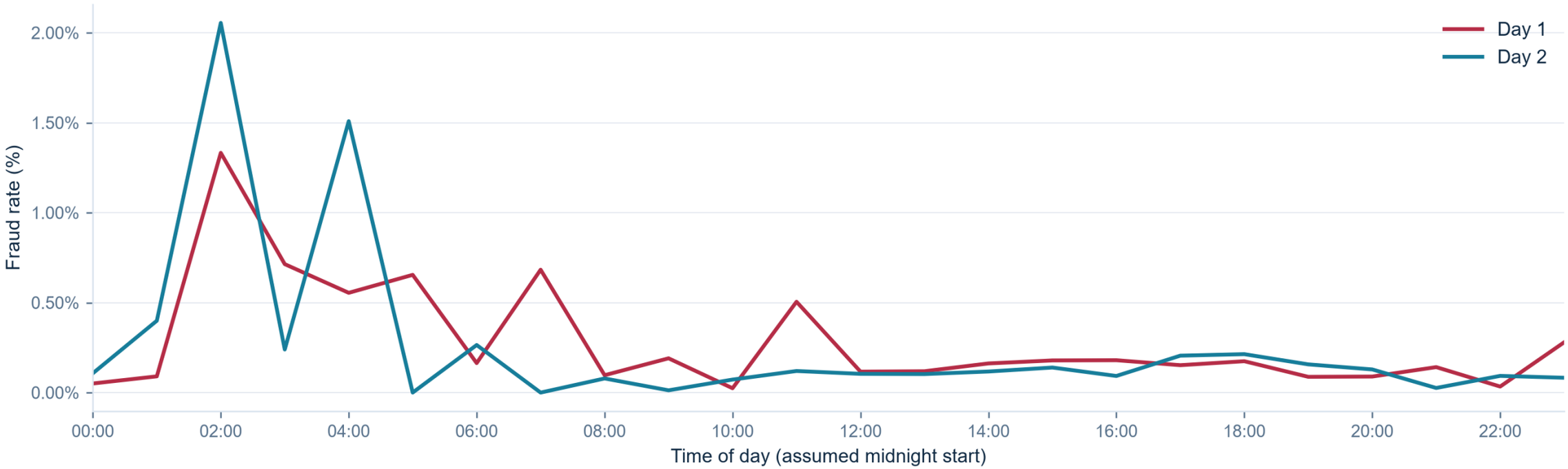
Time assumption: Day 1 starts at 00:00; calendar anchor 01 Jan 2000 is illustrative. No real source date is available.

Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

The same clock hour can look different from day to day

Compare the rate at each hour of the day. Both days begin at 00:00 under the agreed assumption.

Day 1: fraud / all	Day 1 fraud rate	Day 2: fraud / all	Day 2 fraud rate
281 / 144,786	0.194%	211 / 140,021	0.151%



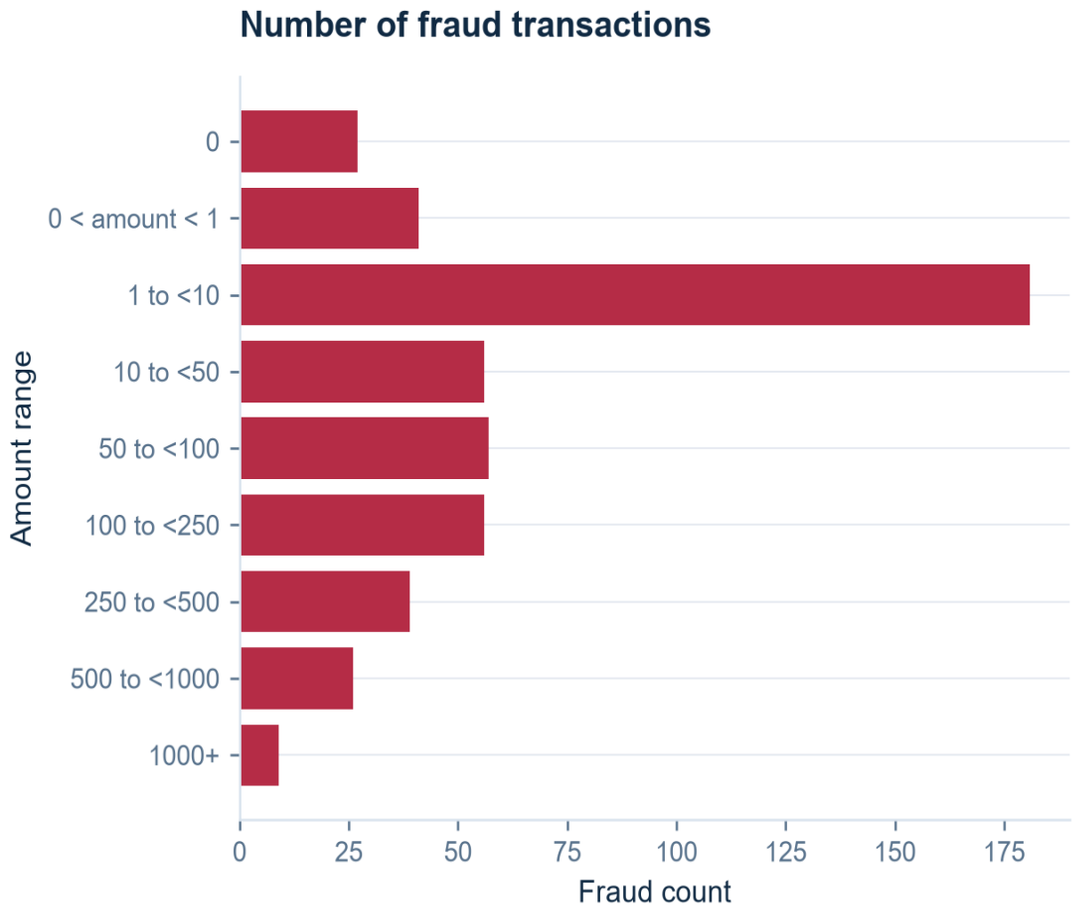
Both days peaked between 02:00 and 03:00. Taken together, that hour contained 57 fraud payments among 3,328 transactions (1.71%). Day 2 also had a second spike at 04:00-05:00: 17 frauds among 1,127 payments (1.51%).

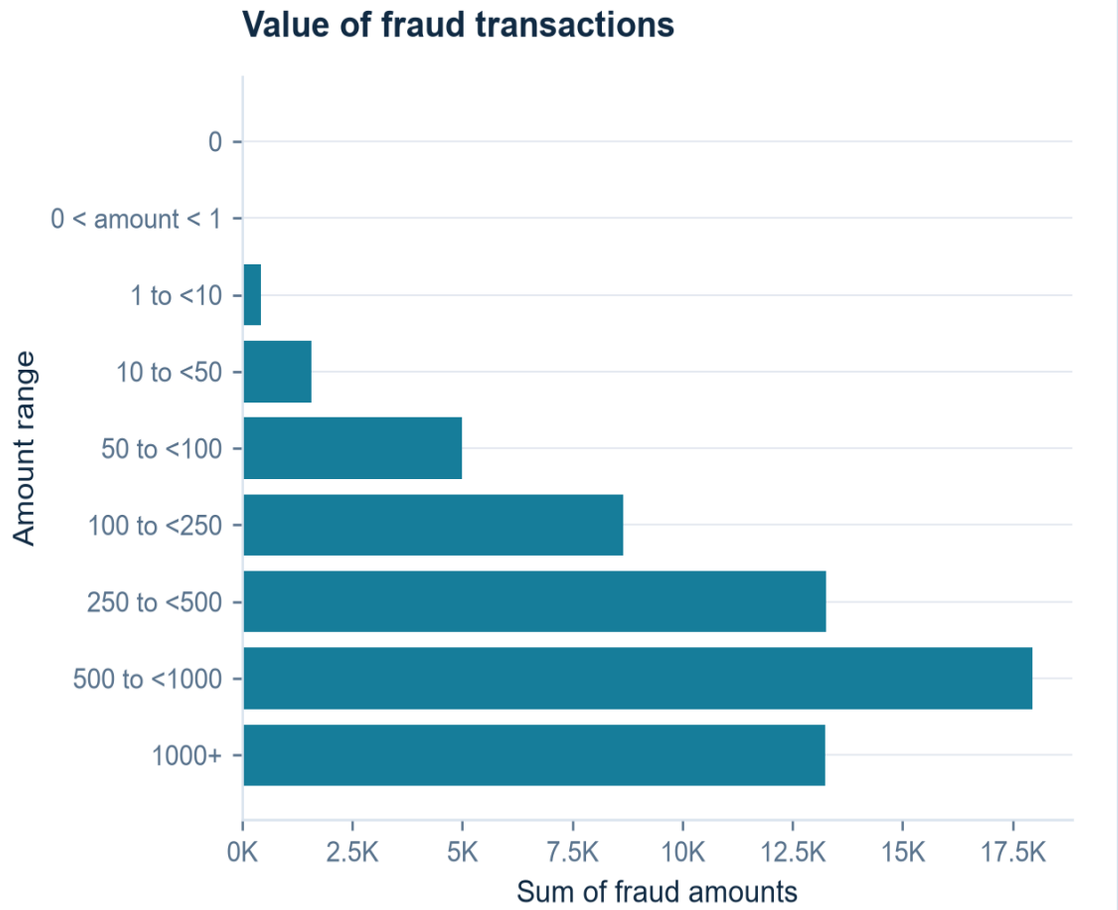
Time assumption: Day 1 starts at 00:00; calendar anchor 01 Jan 2000 is illustrative. No real source date is available.

Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

Small payments dominate the count; large payments dominate value

Count and value answer different questions. Amounts are shown in the units supplied by the file.





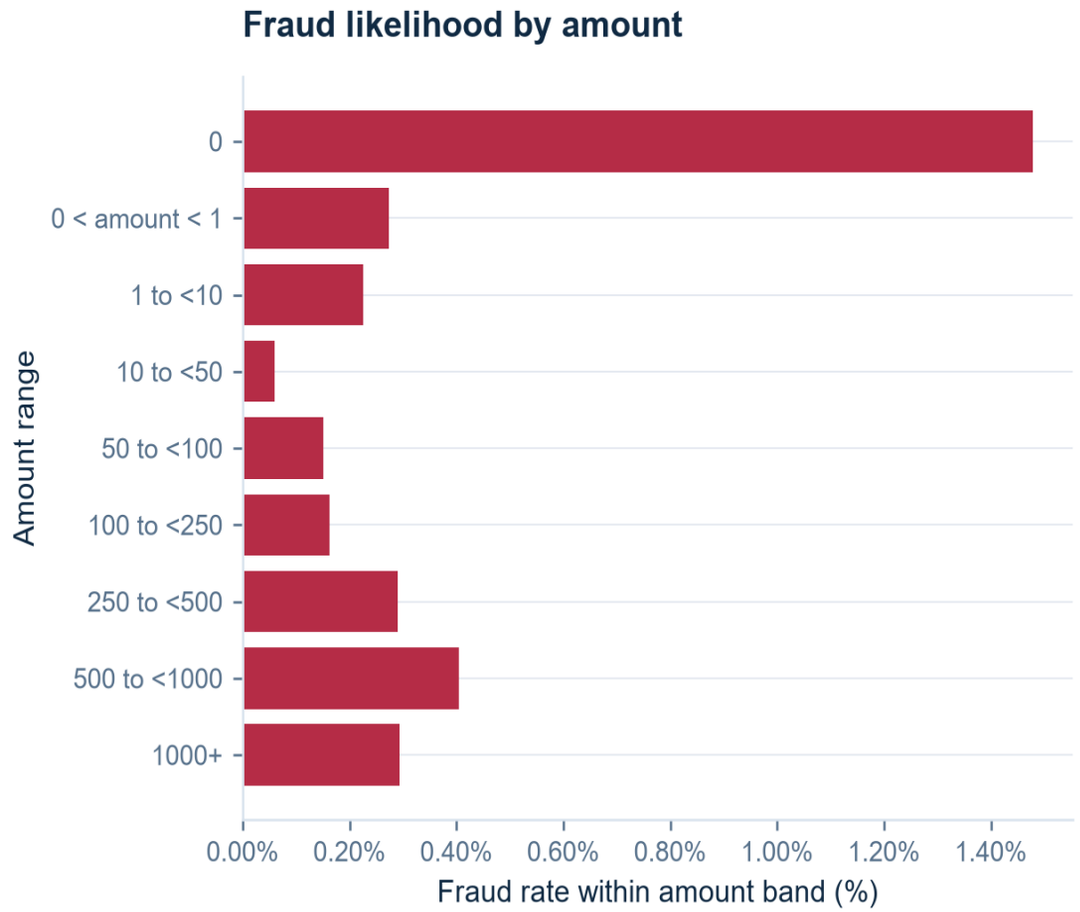
249 fraud payments were below 10 (50.61% of cases), adding up to 458.90. The 74 payments of 250 or more added up to 44,446.57 (73.92% of fraud value). Fraud-labelled amounts are recorded payment value, not a measured net bank loss.

Time assumption: Day 1 starts at 00:00; calendar anchor 01 Jan 2000 is illustrative. No real source date is available.

Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

Which amount bands had the highest fraud rate?

Within-band rate = fraud payments / all payments in the same amount band. Raw records are retained.



Amount band	Fraud	All payments	Fraud rate
0	27	1,825	1.479%
0 < amount < 1	41	14,979	0.274%
1 to <10	181	80,510	0.225%
10 to <50	56	92,390	0.061%
50 to <100	57	37,718	0.151%
100 to <250	56	34,482	0.162%
250 to <500	39	13,411	0.291%
500 to <1000	26	6,423	0.405%
1000+	9	3,069	0.293%

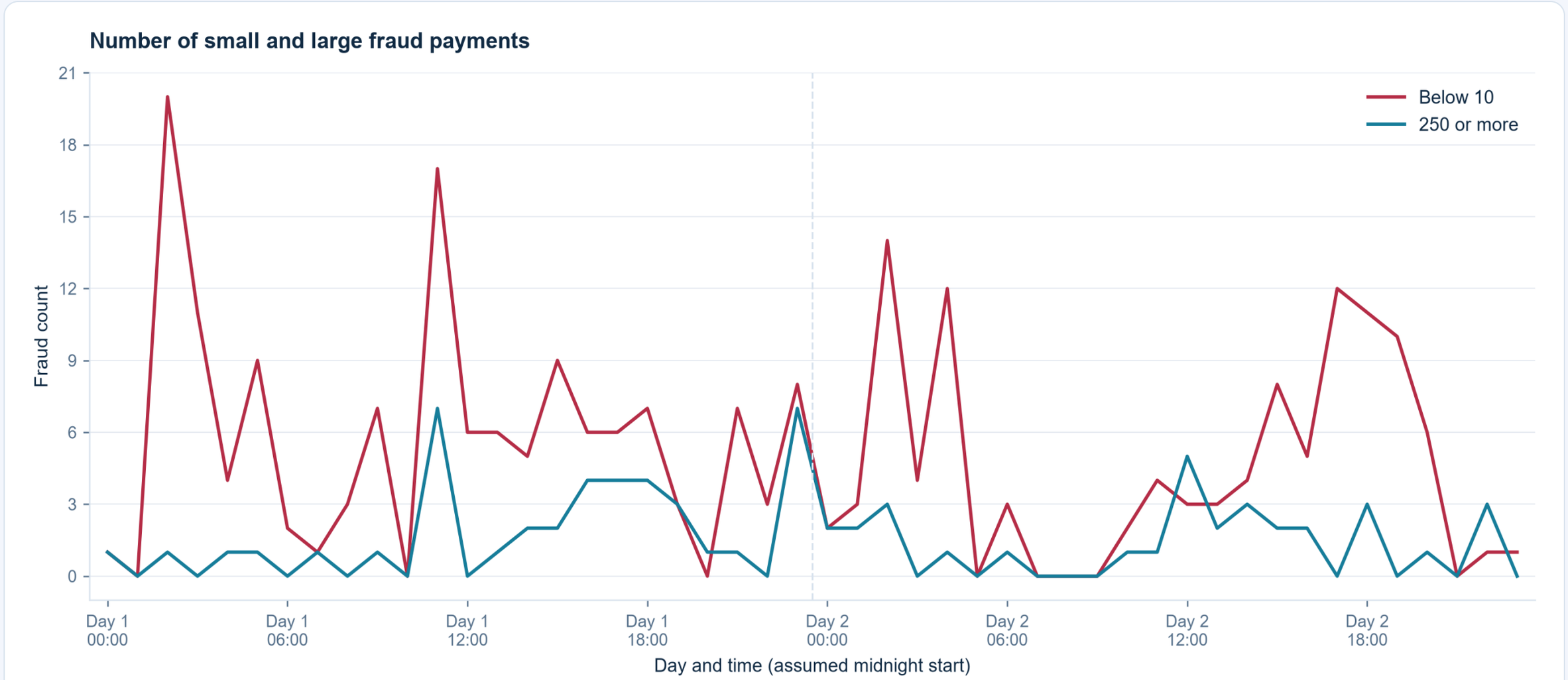
Zero-amount payments had the highest rate: 27 frauds out of 1,825 payments (1.479%). Among positive amounts, the 500 to below 1,000 band had the highest rate (0.405%). This differs from the bands with the most cases or the largest fraud value.

Time assumption: Day 1 starts at 00:00; calendar anchor 01 Jan 2000 is illustrative. No real source date is available.

Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

When did small and large fraud payments happen?

Small = below 10. Large = 250 or more. Counts use the same scale so case volumes can be compared.



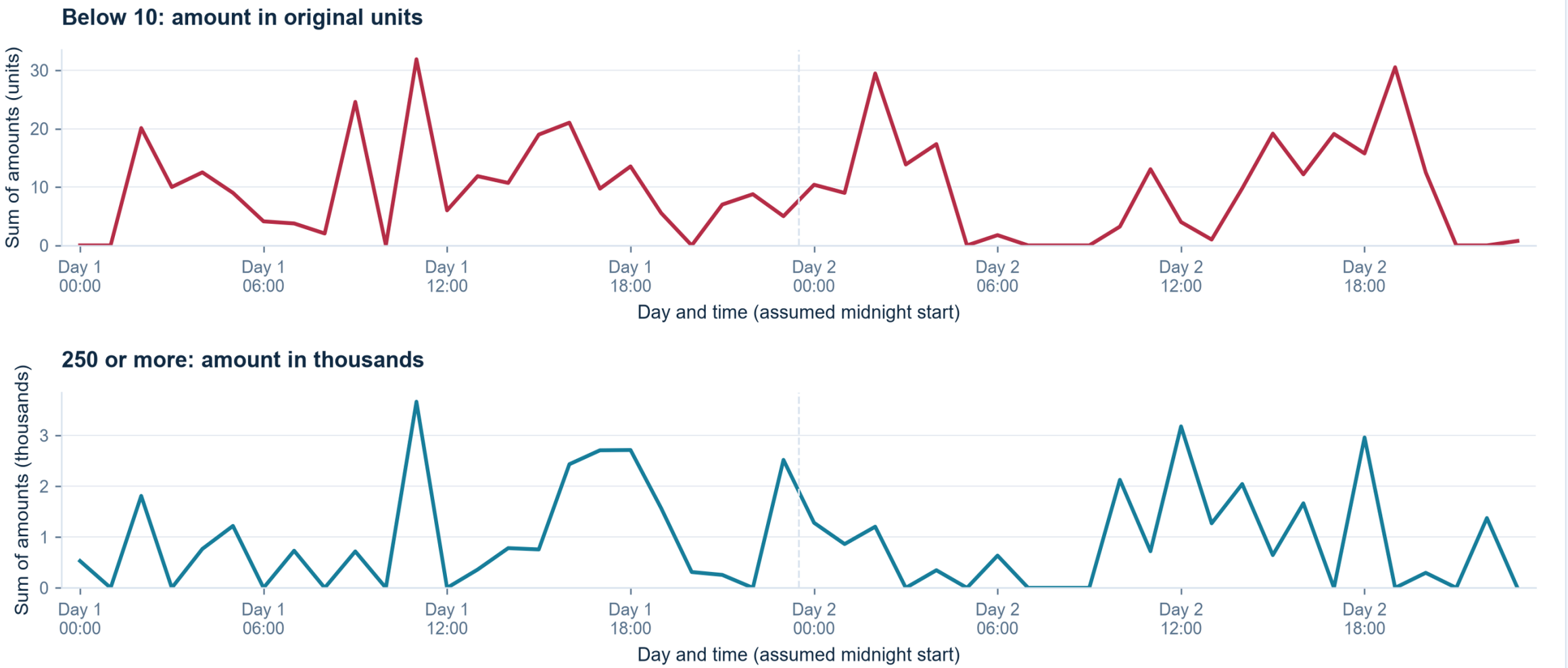
Small payments appear more often across the two-day window. Card IDs are needed to check whether repeated small payments and later large payments belong to the same card. Timing across the full file cannot establish that sequence.

Time assumption: Day 1 starts at 00:00; calendar anchor 01 Jan 2000 is illustrative. No real source date is available.

Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

Small fraud amounts are visible on their own scale

Aligned time axes; separate vertical scales. Compare timing, and read the scale before comparing magnitude.



Small fraud payments totalled 458.90; large ones totalled 44,446.57. Separate scales reveal when small amounts accumulated instead of flattening them against the larger values. Equal line heights on these two panels do not mean equal amounts.

Time assumption: Day 1 starts at 00:00; calendar anchor 01 Jan 2000 is illustrative. No real source date is available.

Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

An amount of 1.00 appeared in 113 fraud payments

Repeated small amounts are visible, but the same amount also occurs in many legitimate transactions.

Fraud payments of 1.00

113

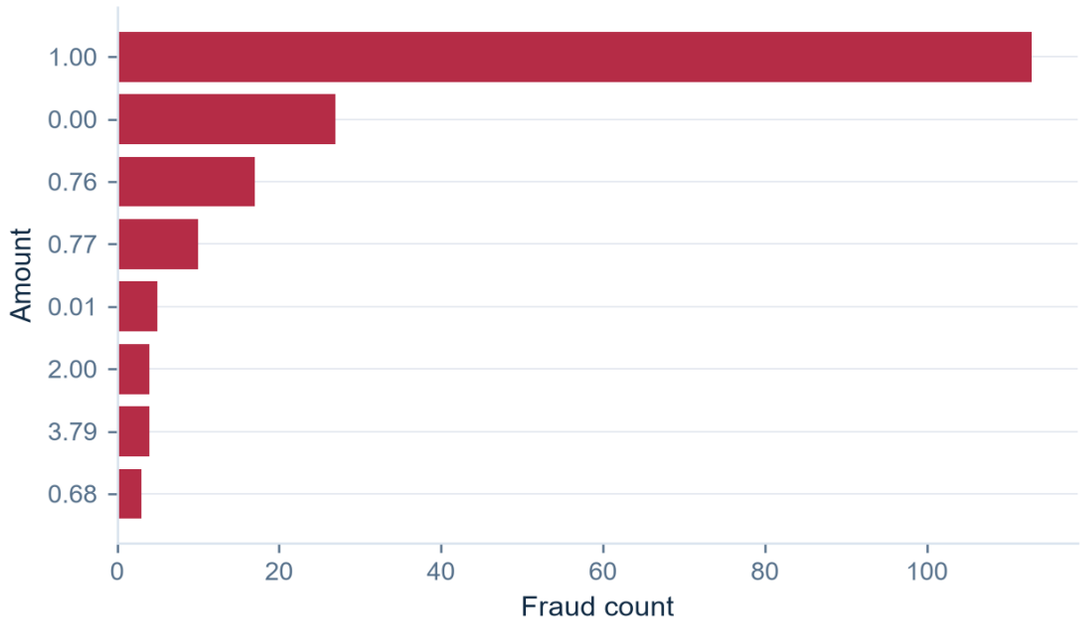
Legitimate payments of 1.00

13,575

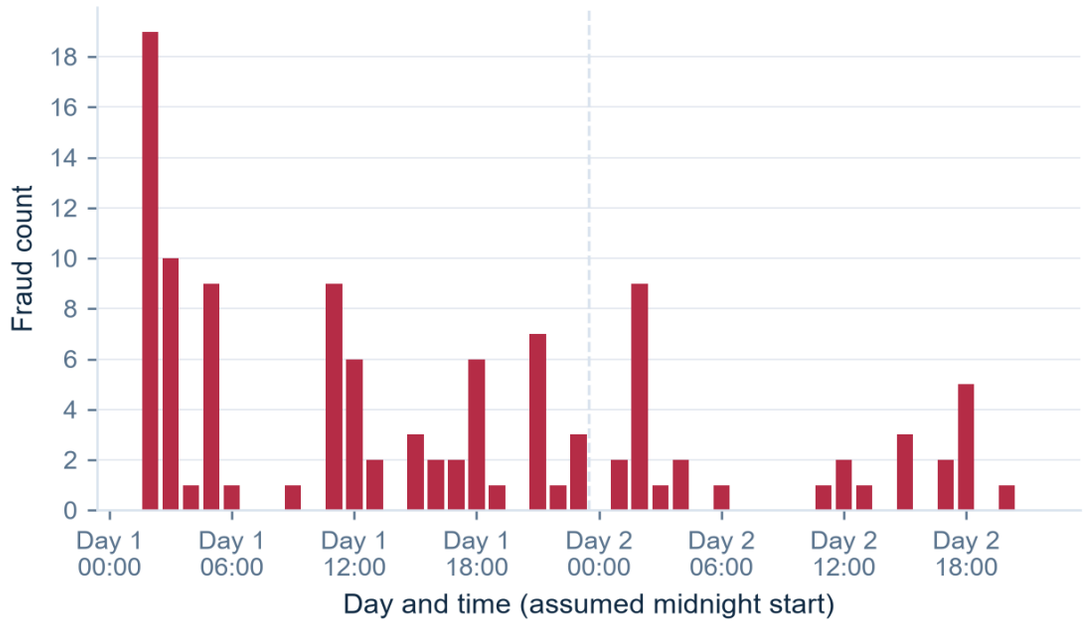
Fraud rate at 1.00

0.826%

Repeated fraud amounts below 10



When the 1.00 fraud payments happened



Of the 113 fraud payments of 1.00, 20 happened from 02:00 to 03:00 on Day 1. The repetition could be worth investigating, but amount alone is a weak rule: 99.17% of all payments of 1.00 were legitimate in this file. Card-level history would be needed to test a card-testing explanation.

Time assumption: Day 1 starts at 00:00; calendar anchor 01 Jan 2000 is illustrative. No real source date is available.

Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

Short waits remain after a repeat-record sensitivity check

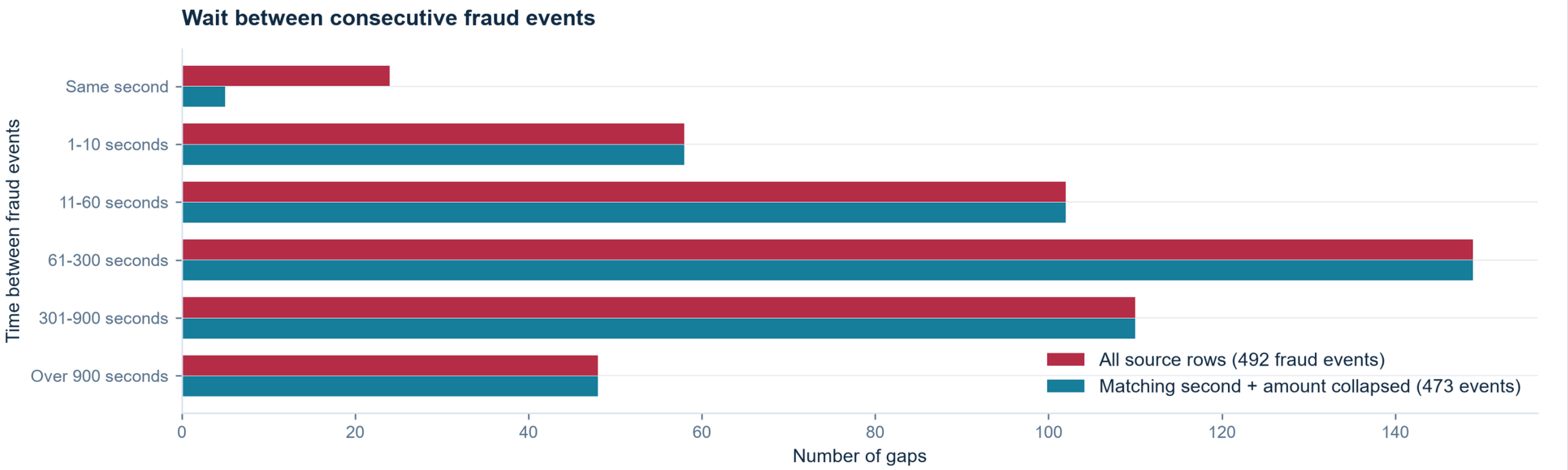
Two views of the fraud stream: all rows, and one row per matching second and amount. This is a sensitivity check only.

Raw: waits of 60 seconds or less

184 / 491 (37.47%)

Collapsed: waits of 60 seconds or less

165 / 472 (34.96%)



Collapsing 19 matching fraud rows reduces same-second gaps from 24 to 5. The median wait rises from 126 to 141 seconds, and the share within one minute falls from 37.47% to 34.96%. The short-wait pattern remains. Matching second and amount may still represent separate real payments, so the main report retains every row.

Time assumption: Day 1 starts at 00:00; calendar anchor 01 Jan 2000 is illustrative. No real source date is available.

Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

Five-minute periods with the most fraud cases

A 300-second window starts at every second. Five non-overlapping periods are selected separately for each ranking.

Rank	Day and time (end excluded)	Fraud	All payments	Fraud rate
1	Day 2 02:01:06-02:06:06	22	155	14.194%
2	Day 1 11:25:14-11:30:14	21	740	2.838%
3	Day 2 04:26:17-04:31:17	10	71	14.085%
4	Day 1 02:04:01-02:09:01	8	175	4.571%
5	Day 1 09:54:14-09:59:14	7	655	1.069%

Top period: fraud / all

22 / 155

Top period: fraud rate

14.194%

The highest count is 22 frauds among 155 payments on Day 2, 02:01:06-02:06:06 (14.194%). Fixed clock buckets had missed part of this burst: their largest count was 20. Full source rows are used in both rolling rankings.

Method: 172,494 complete windows; include start, exclude end. Ties use earliest start. Select the highest remaining window, skip overlaps, then continue to five. No minimum-volume filter is applied. Rankings are a snapshot of the supplied file; re-scan if the source changes.

Time assumption: Day 1 starts at 00:00; calendar anchor 01 Jan 2000 is illustrative. No real source date is available.

Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

Five-minute periods with the highest fraud rates

A 300-second window starts at every second. Five non-overlapping periods are selected separately for each ranking.

Rank	Day and time (end excluded)	Fraud	All payments	Fraud rate
1	Day 2 04:26:50-04:31:50	10	64	15.625%
2	Day 2 02:01:06-02:06:06	22	155	14.194%
3	Day 1 02:23:20-02:28:20	5	89	5.618%
4	Day 2 02:12:19-02:17:19	6	109	5.505%
5	Day 1 04:42:13-04:47:13	3	59	5.085%

Top period: fraud / all

10 / 64

Top period: fraud rate

15.625%

The highest rate is 10 frauds among 64 payments on Day 2, 04:26:50-04:31:50 (15.625%). This differs from the period with the most cases. The denominator is shown because a high rate based on fewer payments is less stable.

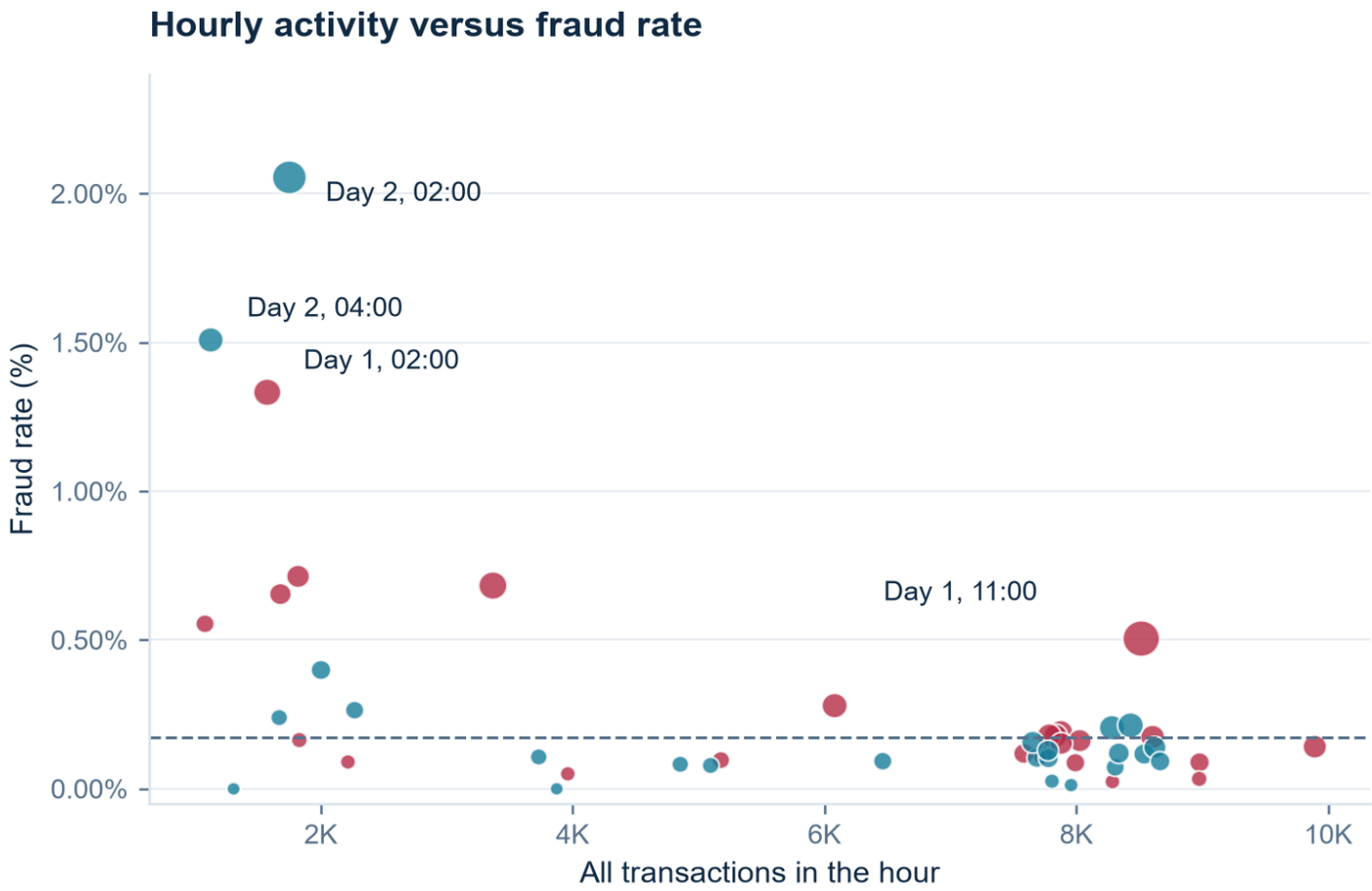
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Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

Review volume and review risk are different decisions

Each point below is one hour. High fraud count means more cases; high fraud rate means a larger share was fraud.



Two queues to investigate

Case workload
Day 1, 11:00-12:00 had 43 fraud payments. This is the hour with the most cases to review.

Fraud concentration
Day 2, 02:00-03:00 had 36 frauds in 1,752 payments. At 2.05%, its fraud rate was about 12 times the whole-file rate.

Marker size
Larger circles mean more fraud cases. Red is Day 1; teal is Day 2. The dashed line is the overall fraud rate (0.173%).

A high rate from a quiet period and a high count from a busy period serve different operational needs. Neither is an automatic blocking rule. The next review should check card-level sequences and whether the timing persists over more days.

Time assumption: Day 1 starts at 00:00; calendar anchor 01 Jan 2000 is illustrative. No real source date is available.

Source: creditcard.csv | Class 1 = confirmed fraud | Amount currency not supplied

What the analysis supports, and what to check next

Use the observed patterns to guide investigation. The file does not contain enough identity or outcome detail to prove motive.

Observed evidence	What to investigate next
02:00-03:00 was the highest-rate hour on both assumed days.	Check more days and the real transaction timezone before allocating a standing overnight review shift.
249 fraud payments were below 10; 113 were exactly 1.00.	Link payments by card and merchant. Check whether small amounts repeat on a card before a larger payment.
74 fraud payments of 250+ carried 73.92% of fraud-labelled value.	Review the large-value cases separately. Obtain actual loss, refund and recovery outcomes before estimating financial impact.
The one-minute gap share falls from 37.47% to 34.96% after matching second and amount are collapsed.	Short waits remain, but check source IDs before calling them an attack burst. Matching time and amount is not proof of a duplicate.
Rolling peaks differ: 22 cases in 155 payments versus a maximum rate of 10 in 64.	Use count to assess case workload and rate to assess concentration. Validate these short windows on more days.

This report uses all 284,807 supplied rows. Class 1 means confirmed fraud; Class 0 means legitimate. The two-day window supports descriptive findings, not a validated fraud-detection model. Original calendar dates, card IDs, merchants, geography and realised losses were not supplied.